

East Java Precipitation Anomalies Across Historical El Niño Episodes (2001–2025)

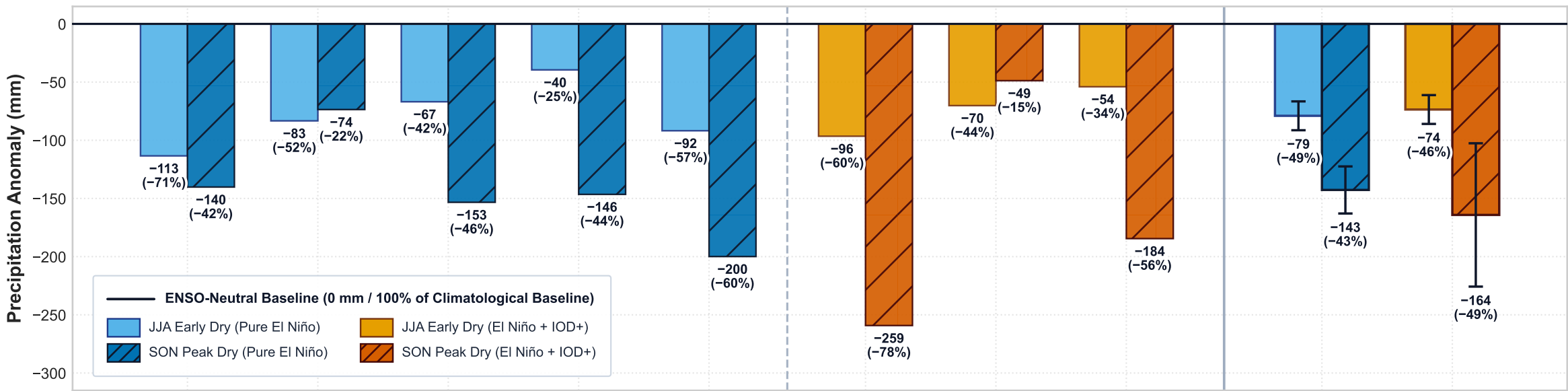
ENSO-Neutral Baseline Climatology (n = 6): JJA = 161 mm, SON = 332 mm | Data: UCSB CHIRPS v2.0 (0.05° resolution)

Pure El Niño (n = 5)

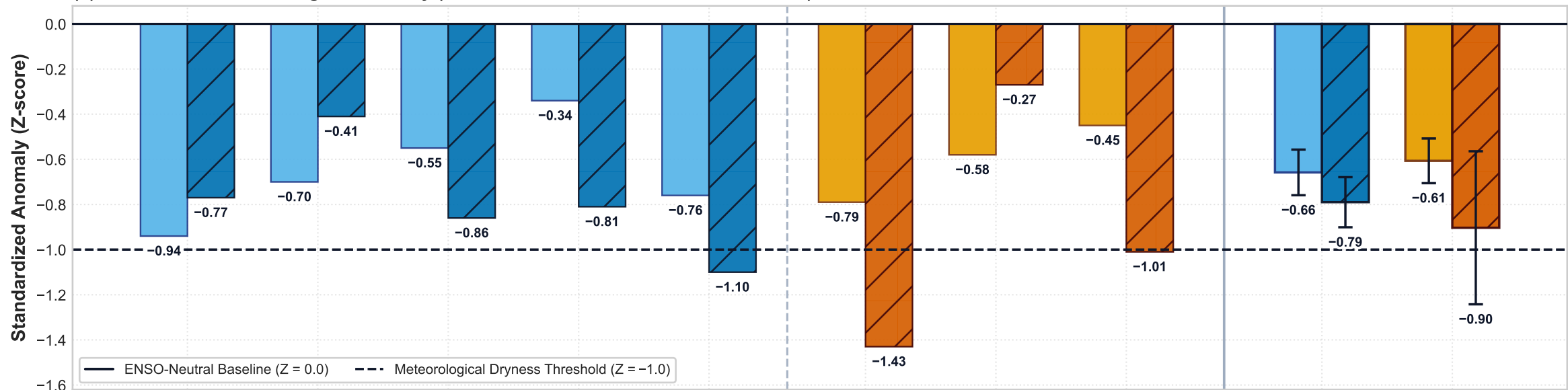
Compound: El Niño + IOD+ (n = 3)

Cohort Composites (±1 SEM)

(a) Seasonal Precipitation Anomaly (mm) and Relative Departure (% of Climatological Baseline)



(b) Standardized Meteorological Anomaly (Z-score relative to ENSO-Neutral Baseline)



Historical El Niño Episodes (2001–2025)
[*2014 & 2015 indicate consecutive dry seasons of the extended 2014–2016 multi-year ENSO episode]

Cohort Composites
[Mean ± 1 SEM]

• Data Source: UCSB CHIRPS v2.0 (0.05° resolution blended precipitation archive).
• Forcing Criteria: Pure El Niño (NOAA ONI ≥ +0.5°C); Compound (+ BoM DMI ≥ +0.4°C).
• Temporal Continuity: *2014 & 2015 denote consecutive dry seasons of multi-year event.

• Neutral Baseline: 6 ENSO-neutral years (JJA = 161 mm ± 49 SEM, SON = 332 mm ± 74 SEM).
• Relative Departure: % loss = (Anomaly / Baseline) × 100%. Zero line (y = 0) = 100% normal.
• Statistical Rigor: Error bars ±1 SEM. Cohort diff p = 0.77 (n.s.); compound drives tail risk.